

Ritul K. Soni

Chicago, IL | (708) 527-9456 | ritulsonicareer@gmail.com | www.linkedin.com/in/ritulsoni | <https://ritulsoni.com/>

EDUCATION

University of Illinois Urbana-Champaign Champaign, IL
Masters of Computer Science May 2025
Bachelor of Science in Computer Science & Economics May 2024
Relevant Coursework: Systems of Generative AI, Deep Learning for Healthcare, Applied ML, Computer Systems

EXPERIENCE

Country Financial July 2024 – Present
AI/ML Engineer Remote

- Deployed company's first **production GenAI system**: ReAct RAG agent built with **LangGraph** serving 500+ insurance agents, reducing average call duration by 40% through retrieval from 5,000+ knowledge articles
- Developed comprehensive **LLM evaluation framework** using LangSmith with Judge-by-LLM metrics for completeness, correctness, and source accuracy, validated against 250+ human-curated ground truth Q&A pairs
- Building a multimodal **document classification system** using OCR, embeddings, and fine-tuned models to automate manual indexing process, targeting 80%+ efficiency improvement for document intake team
- Developing customer-facing **multi-agent LangGraph system** for policy inquiries with multi-document retrieval (pre/post policy states, premium explanations) to automate complex customer service workflows

Country Financial Digitalab August 2023 – July 2024
Software Engineer Intern Champaign, IL

- Prototyped initial RAG chatbot integrating Azure OpenAI and LangChain for eGain knowledge management system retrieval, enabling agents to access internal documents and procedures
- Led research initiative as Project Manager to evaluate GenAI coding assistants, presenting strategic recommendations for technology infrastructure enhancement

NYU - AI Zoning & Regulation Project May 2024 - September 2024
Generative AI Research Assistant Remote

- Enhanced RAG system accuracy through **hybrid search and reranking** techniques, implementing jurisdiction-based metadata and namespace filtering for improved vector database retrieval
- Optimized retrieval pipeline with asynchronous API calls and batching, reducing inference costs by 50% while maintaining accuracy through fine-tuning experiments

Pathlit AI January 2024 – April 2024
Software Engineer Chicago, IL

- Designed and implemented AI-driven backend agents integrating with LLMs, including a versatile API agent, robust web scraper, and precise PDF table extractor for automated data processing
- Pioneered RAG integration with LLMs to streamline Generative AI workflows

RESEARCH & PUBLICATIONS

Reasoning-Enhanced Healthcare Predictions with RL Search Agent & RAG Systems In Progress
Reproducing and Extending Bias Analysis in Clinical Contextual Word Embeddings [pdf][code] 2025
CodeBlend: Efficient KV-Caching Methods for Graph-Based Code Dependencies [pdf][code] 2024
LLM vs. Self-Reported Sentiment: A Comparative Analysis of Happiness Measurement [pdf][code] 2024

TECHNICAL SKILLS

AI/ML Frameworks: LangChain, LangGraph, LangSmith, TensorFlow, PyTorch, Transformers, RLHF/RLAIF
Model Evaluation: Custom benchmark design, failure taxonomy development, adversarial testing, Judge-by-LLM
Languages & Tools: Python, C++, JavaScript/TypeScript, Git, Docker, Azure ML, OpenAI API, OCR, Embeddings
Research Areas: RAG systems, Agentic AI, multimodal reasoning, model safety, automated evaluation
Spoken Languages: English, Spanish, Hindi, Gujarati, Urdu